Notes: Miller (JF, 1977)

Debt and Taxes

Contents

1 Big picture

- **Question.** Does the deductibility of corporate interest payments imply that firms should lever up until bankruptcy costs stop them? Miller's answer is: *not in market equilibrium.* Once personal taxes are brought in, the familiar tax gain from debt can be competed away.
- **Conventional view being challenged.** The then-fashionable trade-off story said: debt creates a large corporate tax shield, bankruptcy costs eventually offset it, and the balance determines each firm's optimal debt ratio. Miller argues that this story gets both sides wrong: the tax gain is overstated, and the bankruptcy-cost counterweight is usually too small to do the work assigned to it.
- **Empirical puzzle motivating the paper.** Corporate debt ratios had remained surprisingly stable over long periods even though corporate tax rates had risen enormously since the 1920s. If the tax advantage of debt were really as large as partial-equilibrium calculations suggested, firms should have moved much more aggressively toward leverage.
- **Main tax result.** The value gain from leverage to stockholders is not simply $\tau_C B_L$. With personal taxes on interest and equity income,

  $$G_L = \left[1 - \frac{(1-\tau_C)(1-\tau_{PS})}{1-\tau_{PB}}\right] B_L.$$

  This collapses to the usual corporate-tax formula only when income from bonds and stock faces the same personal tax treatment.
- **Main equilibrium result.** There can be an *equilibrium aggregate amount of corporate debt* for the economy as a whole, but *no unique optimal debt ratio for any individual firm.* Different firms can choose different leverage policies and still have the same value because each policy is matched to a different investor clientele.
- **Core mechanism.** Bond interest is taxed more heavily at the personal level than equity returns, especially when equity returns are received partly as deferred capital gains. Investors therefore require higher before-tax yields on bonds. That gross-up in bond yields offsets the corporate deductibility of interest.
- **Clientele result.** Low-leverage firms naturally attract investors in high personal tax brackets; high-leverage firms attract tax-exempt or low-tax investors. One clientele is not intrinsically better than another.
- **Who benefits from the corporate tax system?** Not necessarily stockholders of levered firms. In equilibrium, part of the apparent tax advantage can show up as a *bondholders' surplus* accruing to tax-exempt investors and investors in low tax brackets.
- **What happens to bankruptcy costs?** Miller does not deny that bankruptcy and agency costs exist. He argues instead that they are too small, especially for large firms, to be the main explanation for why firms do not issue much more debt.
- **Economic takeaway.** The paper shifts capital-structure analysis from a *firm-by-firm tax shield calculation* to an *economy-wide equilibrium problem.* Once taxes are viewed through prices and investor clienteles, capital structure can again look close to Modigliani-Miller irrelevance.

2 Setup and economic objects (key objects)

2.1 Tax rates and financial claims

- τ_C: the corporate income tax rate.
- τ_{PS}: the personal income tax rate applicable to income from common stock. Miller's point is that this effective rate can be much lower than the statutory ordinary-income rate because equity returns often arrive as capital gains, can be deferred, and at the time could often escape tax entirely if held until death.
- τ_{PB}: the personal income tax rate applicable to interest income from bonds.
- B_L: the market value of the levered firm's debt.
- G_L: the gain from leverage to stockholders relative to an otherwise identical unlevered firm.

Interpretation. The entire paper turns on the fact that corporate taxes are only one layer of the tax system. The relevant object for investors is the *after-tax* return on the final security they hold.

2.2 Investors, tax clienteles, and bond-market variables

- r_0: the equilibrium interest rate on fully tax-exempt bonds, such as state and local government bonds in the stylized example.
- $r_d(B)$: the demand schedule for taxable corporate bonds. It slopes upward because higher and higher interest rates are needed to attract investors from higher tax brackets into the taxable bond market.
- $r_s(B)$: the supply schedule for corporate bonds. In the simplest riskless-bond benchmark, it is horizontal at the corporate-tax gross-up of the tax-exempt rate.
- B^*: the equilibrium aggregate quantity of corporate bonds outstanding.

Interpretation. Miller is not solving one firm's capital-structure problem holding security prices fixed. He is solving for a *market equilibrium* in which the yields on debt and equity adjust endogenously to taxes and investor demand.

2.3 Baseline assumptions doing the heavy lifting

- Taxes are initially treated as proportional, so the algebra is transparent.
- Debt is initially modeled as perpetual and riskless.
- There are no transaction costs, flotation costs, or surveillance costs in the simplest graphical example.
- Tax authorities are assumed to prevent large-scale tax-arbitrage tricks, such as borrowing to hold tax-exempt securities and converting taxes into negative taxes.
- The relevant comparison is between otherwise identical firms holding the same real assets but financing them with different mixes of debt and equity.

Interpretation. These assumptions are not meant to describe the world literally. They isolate the core tax-equilibrium mechanism: the corporate tax shield on interest is offset by the personal-tax disadvantage of interest income.

2.4 Empirical objects in the background

- **Bankruptcy-cost evidence.** Miller leans heavily on Warner's study of railroad bankruptcies to argue that realized direct bankruptcy costs were small relative to the tax shields often invoked in capital-structure discussions.
- **Debt-ratio evidence.** He also uses the stability of aggregate corporate debt ratios from the 1920s through the postwar period as a stylized fact that is hard to reconcile with a large unexploited tax benefit from debt.
- **Security-design evidence.** Income bonds matter in the paper because they appear to preserve tax deductibility while muting classic foreclosure costs. Their rarity is therefore informative.

3 Main argument and equilibrium characterization (all equations + interpretation)

3.1 Why the standard bankruptcy-cost trade-off looks too weak

- The conventional estimate of the tax saving from permanent debt was roughly **50 cents per dollar of debt**. That is a very large number.
- By contrast, the often-cited **20 percent bankruptcy-cost figure** came mainly from evidence on individuals and small businesses, often in liquidation, not from large publicly traded corporations.
- Warner's evidence on 11 railroads showed average direct bankruptcy and reorganization costs of about **5.3 percent** of firm value at filing, with a strong inverse size effect and only **1.7 percent** for the largest railroad. Since these are ex post costs at the time of failure, expected costs at the time debt is issued are much smaller.
- Warner also found that direct bankruptcy costs were only about **1 percent** of firm value *seven years before filing*. Once one multiplies by the probability that bankruptcy actually occurs, the expected deadweight cost becomes smaller still.
- If standard debt contracts really created deadweight costs large enough to offset such a big tax shield, firms and bankers should have been driven toward alternative securities with lower distress costs. Miller's favorite example is the **income bond**, whose interest is paid only when earned and whose nonpayment does not immediately trigger foreclosure.

Interpretation.

- Miller's point is not that bankruptcy costs are zero.
- His point is that the traditional calculation "tax shield minus bankruptcy costs" compares a very large horse to a very small rabbit. That is why he says the trade-off story looks like a horse-and-rabbit stew.

- If the tax gain from leverage were really huge, market participants should have found contractual ways to preserve the tax shield while reducing the supposed distress costs.

3.2 The tax gain from leverage once personal taxes are included

Miller's key expression for the gain from leverage to stockholders is

$$G_L = \left[1 - \frac{(1-\tau_C)(1-\tau_{PS})}{1-\tau_{PB}}\right] B_L. \tag{1}$$

Interpretation of equation (1).

- The familiar corporate-tax formula $\tau_C B_L$ is only a special case.
- The numerator $(1-\tau_C)(1-\tau_{PS})$ captures the after-tax payoff to equity once both corporate and personal taxes are applied.
- The denominator $(1-\tau_{PB})$ captures the personal tax burden on interest income.
- If bondholders must pay more tax on interest than stockholders pay on equity income, the market requires a higher pre-tax bond yield, and that higher yield offsets the firm's interest deduction.

3.3 Special cases and the zero-gain condition

Equation (1) nests several important benchmark cases.

1. **No taxes:**
$$\tau_C = \tau_{PS} = \tau_{PB} = 0 \implies G_L = 0.$$
This reproduces the original Modigliani-Miller no-tax irrelevance result.
2. **Equal personal tax treatment of debt and equity:**
$$\tau_{PS} = \tau_{PB} \implies G_L = \tau_C B_L.$$
This is the standard "corporate tax shield" result.
3. **Debt taxed more heavily than equity:**
$$\tau_{PS} < \tau_{PB} \implies G_L < \tau_C B_L.$$
The tax advantage of debt shrinks and may disappear.

The gain from leverage is exactly zero when

$$1 - \tau_{PB} = (1-\tau_C)(1-\tau_{PS}). \tag{2}$$

Interpretation of equation (2).

- This is the knife-edge case where the corporate tax benefit of deducting interest is exactly offset by the personal-tax disadvantage of receiving interest.
- Miller's larger point is that in equilibrium the system can be driven toward precisely this kind of offset, because prices and ownership patterns adjust.

3.4 Where equation (1) comes from

The paper's footnote derivation uses a standard replication argument.

- Holding a fraction α of a levered firm yields after-tax cash flow proportional to

$$\alpha(\tilde{X} - rB_L)(1 - \tau_C)(1 - \tau_{PS}),$$

  where $\tilde{X}$ is the uncertain return on the real assets.

- This can be replicated by holding equity in the corresponding unlevered firm and borrowing personally in just the right amount. Because personal borrowing interest is deductible, the net borrowing cost involves $(1 - \tau_{PB})$.
- Comparing the two packages yields equation (1).

Interpretation. The formula is not a trick. It is the natural asset-pricing implication of evaluating securities by their *after-tax* payoffs.

3.5 Bond-market equilibrium and Figure 1

Miller then moves from a single-firm comparison to an aggregate market equilibrium. For expositional simplicity he sets the personal tax rate on stock income to zero and assumes riskless bonds.

For an investor in tax bracket α, the required interest rate on taxable corporate bonds is

$$r_d^\alpha = \frac{r_0}{1 - \tau_{PB}^\alpha}. \tag{3}$$

The supply side for corporations is the tax-exempt rate grossed up by the corporate tax rate,

$$r_s(B) = \frac{r_0}{1 - \tau_C}. \tag{4}$$

The intersection of the upward-sloping demand schedule and this horizontal supply schedule determines the aggregate equilibrium quantity of corporate debt,

$$B = B^*. \tag{5}$$

Interpretation of equations (3)–(5).

- The demand curve slopes upward because progressively higher yields are needed to attract investors in progressively higher tax brackets into taxable bonds.
- The supply line is flat in the simplest benchmark because, absent default and other frictions, corporations are indifferent at the rate that exactly converts tax-exempt borrowing into taxable borrowing after the corporate deduction.
- If the market offers *more* debt than B^*, interest rates rise above the level at which debt is attractive, so some firms delever.
- If the market offers *less* debt than B^*, interest rates are too low, so some currently unlevered firms find debt attractive and issue more.

The key conclusion is:

- there is an equilibrium *aggregate* debt ratio for the corporate sector;
- there is *no unique optimal debt ratio* for an individual firm.

A firm like IBM or Kodak can choose low leverage and appeal to investors in high tax brackets. A regulated utility can choose high leverage and appeal to low-tax or tax-exempt investors. In equilibrium, *one clientele is as good as the other*.

3.6 Bondholders' surplus and who captures the tax benefit

- Once the marginal bondholder's personal tax rate is driven up to the relevant equilibrium level, taxable bond yields reflect that tax burden.
- Tax-exempt institutions and low-bracket investors then earn a **bondholders' surplus**: they receive yields set to compensate the marginal taxable investor, even though their own tax burden is lower.
- This means the corporate tax system can transfer value to bondholders rather than to stockholders.
- Miller also stresses that this cuts both ways: low-bracket individuals and corporations effectively bear the corporate tax when *they* borrow.

Interpretation. The right question is not "how much tax does the firm save by deducting interest?" The right question is "who captures the price effects once all investors and all securities re-equilibrate?"

3.7 Why the effective tax rate on stock income may be very low

- Stocks themselves have clienteles: high-dividend stocks are naturally held by tax-exempt organizations and lower-income investors, while low-dividend, high-capital-gain stocks are attractive to higher-bracket investors.
- Capital gains taxation is especially favorable because taxes can be deferred until realization.
- At the time of the paper, holding until death often meant that gains could avoid personal tax entirely because of the step-up in basis.

Interpretation. This is why Miller treats the assumption $\tau_{PS} \approx 0$ as a useful simplification rather than an absurdity. The paper needs equity income to be taxed *more lightly than interest*, not necessarily literally at zero.

3.8 What changes once debt is risky

- Default risk complicates the clean benchmark because firms in distress may not be able to use the full interest deduction. Some tax shields are wasted when taxable income is too low.
- In that case, the relevant risk-adjusted supply rate for debt lies *below* $r_0/(1 - \tau_C)$, and the reduction is larger when default probability is higher.

- Bankruptcy costs and lending costs push in the same direction.
- Importantly, part of those costs is shifted to bondholders in equilibrium through lower risk-adjusted yields, not borne entirely by stockholders.

Interpretation. Risk, default, and bankruptcy do matter at the margin, but they do not restore the simple idea that each firm has a large private tax gain from debt waiting to be harvested.

3.9 Methodological point: firms need not literally solve the model

- Miller directly addresses the objection that "firms and investors do not behave that way."
- He agrees that real decision making is heuristic, judgmental, and imitative, not the literal maximization problem of the model.
- His defense is evolutionary: heuristics compatible with rational market equilibrium survive, while harmful ones are selected out.
- Neutral heuristics can persist for long periods even if they have no deep optimizing rationale, and they may become valuable when the environment changes.

Interpretation. The equilibrium model is not a psychological description of the CFO's brain. It is a market-level selection argument about which financing patterns can persist.

4 What makes the argument credible (and where it is limited)

4.1 What does the heavy lifting

- **Personal-tax asymmetry.** Interest income is generally taxed more heavily than equity income.
- **Investor heterogeneity.** Investors sit in different tax brackets, and some are tax exempt.
- **Clienteles.** Securities sort into the hands of investors who value them most on an after-tax basis.
- **General equilibrium rather than partial equilibrium.** Debt and equity yields are not fixed; they adjust until arbitrage opportunities vanish.

4.2 Why the empirical record supports the question Miller is asking

- The typical nonfinancial corporation's debt ratio in the 1950s looked surprisingly similar to that of the 1920s even though the corporate tax rate had increased from roughly 10–11 percent to about 52 percent.
- Even if one interprets the 1960s as a real rise in leverage rather than partly an accounting artifact, the move is modest relative to what the conventional tax-shield calculation would predict.
- The much-discussed leverage spike of the mid-1970s is presented as largely cyclical and temporary, tied to inventory accumulation and short-term borrowing during a peculiar macroeconomic episode.

Interpretation. Miller is not claiming that leverage never changes. He is arguing that the time-series behavior of debt ratios looks far too muted to be driven by a large unexploited tax subsidy to debt.

4.3 What the paper proves and what it does not

- The paper proves that with corporate and personal taxes, a market equilibrium can exist in which firm value is still independent of capital structure.
- It does *not* prove that bankruptcy costs, agency costs, or contracting frictions are irrelevant in every context.
- It does *not* build a fully detailed cross-sectional model explaining every industry's debt ratio.
- It does *not* offer a literal description of how actual treasurers compute financing decisions.

4.4 Main limitations

- **Tax simplification.** The clean formulas rely on proportional-tax logic and stylized tax treatment of equity and debt.
- **Benchmark debt is riskless.** The clearest version of Figure 1 abstracts from default risk, wasted tax shields, and distress frictions.
- **Clientele evidence is indirect.** The paper's equilibrium mechanism depends on tax sorting by investors, but the direct empirical evidence on that sorting was limited at the time.
- **Firm-level frictions are downplayed.** Agency costs, information asymmetry, regulation, collateral, and maturity structure are not central in the benchmark model.

4.5 Relation to the literature

- Relative to Modigliani-Miller (1958, 1963), the paper is a major tax-equilibrium extension.
- Relative to the emerging static trade-off theory, it argues that the relevant tax comparison must include *personal* taxes and market-clearing yields.
- Relative to later corporate-finance research, the paper is one of the canonical foundations of the "Miller equilibrium" view: tax clienteles can rationalize aggregate leverage without firm-level optimal leverage targets driven solely by tax shields.

5 Figure 1 and key empirical facts

5.1 Section I: bankruptcy costs in perspective

Read.

- Miller emphasizes the gap between the commonly cited tax gain from debt and the much smaller direct bankruptcy-cost evidence for large firms.

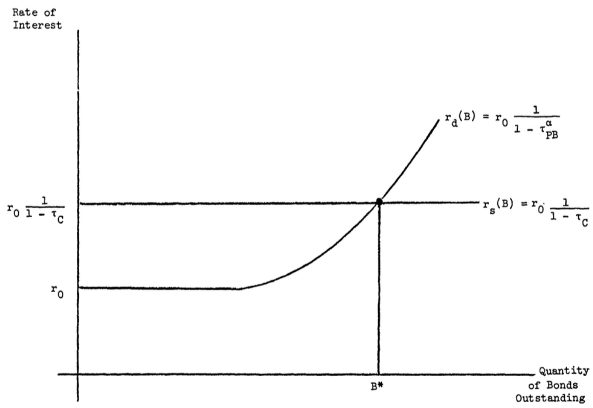

FIGURE 1. Equilibrium in the Market for Bonds

- He uses Warner's railroad evidence to show that direct costs at failure are only a few percent of firm value and expected costs ex ante are much smaller.
- He also points to income bonds as a security that appears to preserve tax deductibility while sharply reducing ordinary foreclosure risk.

Why it matters.

- This section weakens the standard counterweight in trade-off theory.
- It sets up the need for a different explanation of why firms are not much more levered.

5.2 Section II: the empirical record on leverage

Read.

- The representative nonfinancial corporation's debt ratio changes surprisingly little across eras with very different tax rates.
- The 1960s rise in leverage is modest and partly confounded by accounting changes such as accelerated depreciation and other tax-rule changes.
- The early-1970s debt scare is interpreted as a temporary inventory and short-term-finance episode, not a structural reoptimization of capital structures.

Why it matters.

- If firms had been sitting on a huge unexploited tax gain from debt, one would expect a much larger and more persistent leverage response.
- This empirical stability is one of the paper's strongest motivating facts.

5.3 Figure 1: equilibrium in the market for bonds

Read.

- The vertical axis is the interest rate; the horizontal axis is the quantity of corporate bonds outstanding.

- The demand curve $r_d(B)$ begins at the tax-exempt rate r_0 and slopes upward because progressively more heavily taxed investors require progressively higher yields to hold taxable bonds.
- The supply curve $r_s(B)$ is horizontal at $r_0/(1-\tau_C)$ in the simple benchmark.
- Their intersection determines B^*, the equilibrium aggregate amount of corporate debt.

Why it matters.

- This figure is the paper's core visual argument.
- It shows immediately why there can be an equilibrium debt ratio for the corporate sector as a whole without there being any unique optimal leverage ratio for any one firm.
- It also makes clear that tax effects show up through *prices and ownership patterns*, not just through a mechanical increase in firm value from issuing debt.

5.4 Section V: market equilibrium and actual behavior

Read.

- Miller explicitly concedes that firms do not literally solve the model's maximization problem.
- He argues instead that market equilibrium is a selection result: harmful financial heuristics tend to die out, while neutral or approximately rational heuristics can persist.

Why it matters.

- This is the paper's methodological defense of finance theory.
- It explains why equilibrium reasoning can still be useful even when actual corporate decision making is messy and heuristic.

6 Short summary

- **Method.** Miller reanalyzes capital structure in a market-equilibrium framework that includes both corporate and personal taxes.
- **Main conceptual result.** The corporate tax advantage of debt is not generally $\tau_C B_L$; it is reduced, eliminated, or even reversed once the heavier personal taxation of interest income is recognized.
- **Main equilibrium result.** The economy can have an equilibrium aggregate quantity of corporate debt without any individual firm having a unique tax-driven optimal leverage ratio.
- **Empirical lesson.** Small observed bankruptcy costs and stable debt ratios are hard to reconcile with a large unexploited tax shield from debt.
- **Big takeaway.** Capital structure has to be analyzed in general equilibrium, with investor clienteles and security prices adjusting endogenously to taxes.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the deductibility of interest at the corporate level does *not* by itself imply that firms should load up on debt. Once personal taxes are incorporated and markets are allowed to clear, the apparent tax advantage of leverage can be competed away. In that sense, firm value can remain independent of capital structure even in a taxed economy.

7.2 Economic intuition

Debt looks attractive in partial equilibrium because the firm deducts interest. But the investors who receive that interest usually pay more personal tax on it than they would on equity returns. The market therefore grosses up the before-tax yield on bonds. That gross-up offsets the firm's deduction. The result is an economy-wide clientele equilibrium: some firms issue more debt, some less, and all can be equally valued.

7.3 Takeaway

Miller's contribution is to move the debt-and-taxes question from the corporate treasurer's worksheet to the equilibrium of the whole capital market. The paper is foundational because it shows that taxes matter through security prices, investor sorting, and market clearing—not just through a firm-level tax shield calculation.